

Devesh Kumar Upadhyay

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Summary

Junior Software Developer with 8+ months of experience building production-grade real-time systems in the healthcare domain. Developed a Digital ICU Management System integrating medical devices via HL7 and TCP/IP, implementing real-time ECG waveform streaming with 25-45x compression. Built an AI-powered bilingual voice assistant (Lucy) using LLM, STT, and TTS for hospital call automation. Strong in ASP.NET Core, C#, EF Core, SignalR, and SQL Server.

EDUCATION

Chandra Shekhar Azad University of Agriculture & Technology

B. Tech in Computer Science & Engineering (OGPA - 8.09)

Kanpur, India

June 2020 - July 2024

EXPERIENCE

Jr. Software Developer

Chhatrapati Shivaji, Subharti Hospital

Aug 2025 - Present

Meerut, UP

- Reduced ECG waveform storage costs by 25–45x in SQL Server by engineering a GZip + delta encoding compression pipeline, processing 1,000+ samples/sec from live patient monitors via SignalR and Canvas rendering.
- Eliminated EF Core DbContext concurrency crashes affecting parallel ECG decompression by architecting a chunked loading system using IServiceScopeFactory for safe scope isolation.
- Cut medical device integration time by designing a unified HL7 + TCP/IP parsing layer that normalized data from two hardware vendors (Mindray & Comen), enabling structured communication for ventilators and patient monitors.
- Reduced corrupted patient record incidents to near-zero by building a UHID normalization service that auto-corrected staff data-entry errors at the point of ingestion.
- Improved ICU data retention compliance by implementing a 7-day automated purge strategy with IST-aware cutoff logic, preventing unbounded database growth in a production healthcare environment.

Apprentice Engineer

366Pi Technologies

Oct 2024- Jan 2025

Ranchi, India

- Improved regression test coverage for web and mobile applications by building automated test suites using Selenium (C# / ASP.NET Core) and Appium across functional and data-driven scenarios.
- Reduced query execution time on slow reports by performing SQL query optimization, including index analysis and execution plan review.

PROJECTS

Digital ICU Management System | ASP.NET Core, C#, SQL, HL7, REST APIs, SignalR - [DemoLink](#)

- Developed a real-time ICU monitoring system integrating medical devices like ventilators and patient monitors.
- Built backend services to receive, parse, and validate device data over TCP/IP.
- Implemented HL7-based data processing for structured medical communication.
- Designed scalable backend using ASP.NET Core and SignalR for real-time updates.
- Optimized database design using SQL Server for efficient storage of patient records and device logs.
- Implemented 7-day data purge strategy and GZip+delta compression achieving 25–45x storage reduction for ECG waveform data

AI-Enabled Hospital Voice Assistant (Lucy) | Python, Asterisk PBX, Groq API (LLaMA 3.3), Whisper STT, Piper TTS, SQL - [DemoLink](#)

- Built a real-time AI voice assistant to handle inbound hospital calls using Asterisk PBX.
- Implemented speech-to-text (Whisper) → LLM reasoning → text-to-speech (Piper) pipeline for natural conversation.
- Enabled bilingual interaction (Hindi/English) for better accessibility.
- Designed intelligent medicine ordering system with database querying, fuzzy matching, and order confirmation.
- Developed end-to-end call automation system with silence detection, turn management, and fallback handling.

SKILLS

Languages:

C#, SQL, HTML/CSS, Python

Databases:

SQL Server, MySQL, Database Design

Tools:

Visual Studio, VS Code, Git, Postman, Swagger

Frontend Technologies:

HTML, CSS, Bootstrap, JavaScript

Backend & Web Technologies:

ASP.NET Core, ASP.NET MVC, Web APIs (REST), Entity Framework Core

Data Analytics:

Power BI, Excel, Exploratory Data Analysis (EDA), Dashboard Development

Soft Skills:

Problem Solving, Clear Communication, Requirement Analysis